



Problem solving with standard form

1 Why is 46×10^{-20} not written in standard form? Tick **1** correct answer

- ☐ The exponent cannot be negative
- ☐ It should be $\div$ and not $\times$
- ☐ 46 is not greater than or equal to 1 and less than 10
- ☐ An exponent cannot be over 10

2 Which of the following are not written in standard form? Tick **2** correct answers

- ☐ 4.6×10^2
- ☐ 0.15×10^{-2}
- ☐ 2×10^{-12}
- ☐ 10.4×10^3
- ☐ 5.006×10^8

3 Jacob writes the number 450×10^3 in standard form. What should Jacob write for the exponent? Fill in the blank

4 Laura writes the number 0.087×10^6 in standard form. What should Laura write for the exponent? Fill in the blank

5 The width of a human hair is 8×10^{-5} metres. The width of 1000 hairs in centimetres is _____ cm. Fill in the blank

6 The population of the world in 2024 is roughly 8 billion. The population of Europe is roughly 740 million. To the nearest integer, the population of the world is _____ times bigger than Europe. Fill in the blank

